Life Satisfaction Prediction using GDP per Capita

Assignment Note

This assignment is not intended to be a full project. It serves as a follow up to reinforce the concepts we have covered in lecture so far. More comprehensive analyses and detailed projects will be introduced as we progress further into the course

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🤔 📚 Learning Objectives

- Load and visualize real-world data
- Train and compare regression models using Scikit-Learn.
- Interpret model outputs and discuss generalization.

In [32]:
```
from sklearn.linear_model import LinearRegression
from sklearn.neighbors import KNeighborsRegressor
```

Part 1: Data Exploration and Visualization (20 marks)

Q1 (5 marks) Load the dataset `lifesat.csv` and display the first 5 rows.

In [12]:
```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

df = pd.read_csv("lifesat.csv")
df.head()
```

Out[12]:

	Country	GDP per capita (USD)	Life satisfaction
0	Russia	26456.387938	5.8
1	Greece	27287.083401	5.4
2	Turkey	28384.987785	5.5
3	Latvia	29932.493910	5.9
4	Hungary	31007.768407	5.6

Q2 (5 marks) Print basic `info` and `summary statistics` .

In [8]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 27 entries, 0 to 26
Data columns (total 3 columns):
 #   Column                Non-Null Count  Dtype  
---  ------                --------------  -----  
 0   Country               27 non-null     str    
 1   GDP per capita (USD)  27 non-null     float64
 2   Life satisfaction     27 non-null     float64
dtypes: float64(2), str(1)
memory usage: 983.0 bytes
```

In [9]:
```
print(df.describe())
```

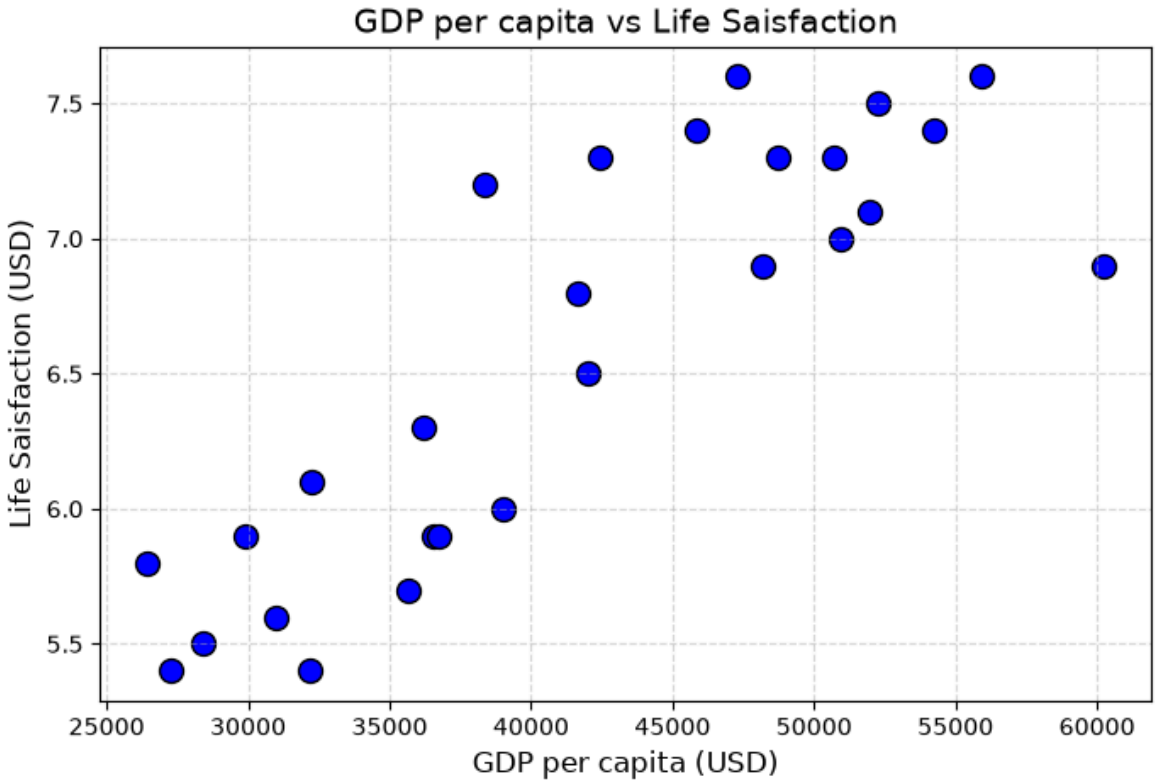
```
       GDP per capita (USD)  Life satisfaction
count             27.000000          27.000000
mean           41564.521771           6.566667
std             9631.452319           0.765607
min            26456.387938           5.400000
25%            33938.289305           5.900000
50%            41627.129269           6.800000
75%            49690.580269           7.300000
max            60235.728492           7.600000
```

Q3 (10 marks) Display a scatter plot for `GDP per capita` vs `Life Satisfaction` . Add labels, title, and discuss the observed relationship.

In [13]:
```
import matplotlib.pyplot as plt

plt.figure(figsize=(8,5))
plt.scatter(df['GDP per capita (USD)'],df['Life satisfaction'],color = 'blue',ed
plt.title('GDP per capita vs Life Saisfaction',fontsize=14)
plt.xlabel('GDP per capita (USD)',fontsize=12)
plt.ylabel('Life Saisfaction (USD)',fontsize=12)

plt.grid(True,linestyle='--',alpha=0.5)
plt.show()
```



Discuss the observed relationship:

There is a positive, roughly linear connection between GDP per capita and life satisfaction —countries with higher GDP also exhibit higher levels of life satisfaction. However, this relationship is not strict; for instance, despite the USA having a high GDP, life satisfaction is lower than expected, indicating that GDP is not the sole determining factor.

Part 2: Linear Regression Model (30 marks)

Q4 (5 marks) Extract input (X) and target (y). Print their shapes.

In [13]:
```
X = df[['GDP per capita (USD)']]
y = df['Life satisfaction']

print(X.shape)
print(y.shape)
```

```
(27, 1)
(27,)
```

Q5 (10 marks) Train a Linear Regression model & Display coefficient and intercept.

In [14]:
```
# To instantiate the model, you need to use LinearRegression()
# use fit function to train the model
from sklearn.linear_model import LinearRegression

lin_reg = LinearRegression()
lin_reg.fit(X,y)

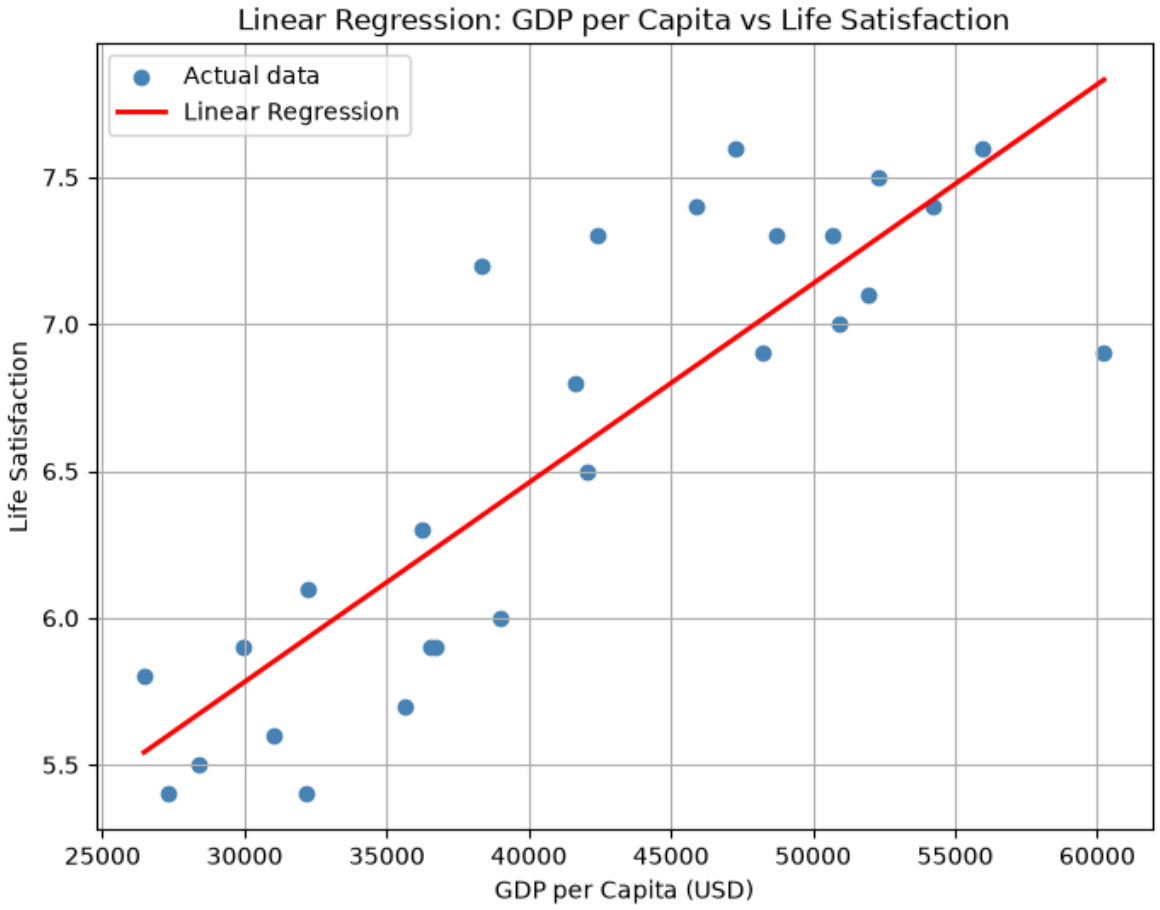
print("Coefficient (slope):", lin_reg.coef_[0])
print("Intercept:", lin_reg.intercept_)
```

```
Coefficient (slope): 6.778899694341222e-05
Intercept: 3.7490494273769093
```

Q6 (10 marks) Plot the predicted regression line from the model along with a scatter plot of the data.

In [17]:
```
import numpy as np
lin_reg.fit(X.values,y)
X_line = np.linspace(X.min(), X.max(), 100).reshape(-1, 1)
y_line = lin_reg.predict(X_line)

plt.figure(figsize=(8,6))
plt.scatter(X, y, color='steelblue', label='Actual data')
plt.plot(X_line, y_line, color='red', linewidth=2, label='Linear Regression')
plt.xlabel('GDP per Capita (USD)')
plt.ylabel('Life Satisfaction')
plt.title('Linear Regression: GDP per Capita vs Life Satisfaction')
plt.legend()
plt.grid(True)
plt.show()
```



Q7 (5 marks) Predict Life Satisfaction for GDP = 37,655.2 USD. Comment on result.

In [18]:
```
gdp_new = [[37655.2]]
lin_reg.fit(X.values,y)
pred_lin = lin_reg.predict(gdp_new)
print(f"Predicted Life Satisfaction for GDP = $37,655.2: {pred_lin[0]:.2f}")
```

```
Predicted Life Satisfaction for GDP = $37,655.2: 6.30
```

Part 3: K-Nearest Neighbors Regression (25 marks)

Q8 (5 marks) Train a KNeighborsRegressor (n_neighbors=3).

In [19]:
```
# you have done this already in the IRIS Flower example
from sklearn.neighbors import KNeighborsRegressor

knn_reg = KNeighborsRegressor(n_neighbors=3)
knn_reg.fit(X, y)
```

Out[19]:

▾ KNeighborsRegressor

▸ Parameters

▸ Fitted attributes

Q9 (10 marks)

- Predict Life Satisfaction for GDP = 37,655.2 USD
- and compare with Linear Regression.

In [32]:
```
gdp_new = pd.DataFrame([[37655.2]],columns=X.columns)
pred_knn3 = knn_reg.predict(gdp_new)
print(f"KNN (k=3) prediction for GDP = $37,655.2: {pred_knn3[0]:.2f}")
print(f"Linear Regression prediction:{pred_lin[0]:.2f}")
print(f"Difference:{abs(pred_knn3[0] - pred_lin[0]):.2f}")
```

```
KNN (k=3) prediction for GDP = $37,655.2: 6.33
Linear Regression prediction:6.30
Difference:0.03
```

Q10 (10 marks)

- Use n_neighbors 1, 3, 5, and 10 and print the predicted values of life satisfaction in Q10.
- Plot the results using a line plot.

In [31]:
```
n_values = [1, 3, 5, 10]
pred_list = []

for n in n_values:
    knn = KNeighborsRegressor(n_neighbors=n)
    knn.fit(X, y)
    gdp_new_df = pd.DataFrame(gdp_new,columns=X.columns)
    pred = knn.predict(gdp_new_df)[0]
    pred_list.append(pred)
    print(f"n_neighbors = {n:2d} -> Predicted Life Satisfaction = {pred:.2f}")
```

```
n_neighbors =  1 -> Predicted Life Satisfaction = 7.20
n_neighbors =  3 -> Predicted Life Satisfaction = 6.33
n_neighbors =  5 -> Predicted Life Satisfaction = 6.26
n_neighbors = 10 -> Predicted Life Satisfaction = 6.37
```

In [30]:
```
# Plot n vs pred_list
plt.figure(figsize=(8,6))
plt.plot(n_values, pred_list, marker='o', color='darkorange', linewidth=2)
plt.axhline(y=pred_lin[0], color='red', linestyle='--', label=f'Linear Regressio
plt.xlabel('n_neighbors (k)')
plt.ylabel('Predicted Life Satisfaction')
plt.title('KNN Predictions vs Number of Neighbors (GDP = $37,655.2)')
plt.xticks(n_values)
plt.legend()
plt.grid(True)
plt.show()
```

